

**Project Design Phase**  
**Proposed Solution Template**

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
| Date          | 30 June 2026                                                                 |
| Team ID       |                                                                              |
| Project Name  | Plugging into the Future: An Exploration of Electricity Consumption Patterns |
| Maximum Marks | 2 Marks                                                                      |

**Proposed Solution Template:**

Project team shall fill the following information in the proposed solution template.

| S.No. | Parameter                                | Description                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Problem Statement (Problem to be solved) | Electricity consumption patterns vary across states, regions, months, and years. There is a need to analyze this data to identify trends, seasonal variations, regional differences, and consumption changes before and after the lockdown for better energy planning and decision-making. |
| 2.    | Idea / Solution description              | Develop an interactive Tableau dashboard to visualize electricity consumption using charts, maps, filters, and dashboards. The solution provides insights into state-wise, region-wise, monthly, quarterly, and yearly electricity usage to support data-driven decisions.                 |
| 3.    | Novelty / Uniqueness                     | Combines multiple analyses such as state-wise, region-wise, Top N & Bottom N, quarter-wise, metro city usage, and before/after lockdown comparison in a single interactive dashboard with dynamic filters.                                                                                 |
| 4.    | Social Impact / Customer Satisfaction    | Helps government agencies, electricity boards, researches, and consumers understand energy usage patterns, improve energy conservation, optimize resource allocation, and support sustainable electricity management.                                                                      |
| 5.    | Business Model (Revenue Model)           | The dashboard can be offered as a Business Intelligence solution to power distribution companies, government departments, and energy consultants through subscription-based analytics or customized reporting services.                                                                    |
| 6.    | Scalability of the Solution              | The solution can be expanded by integrating live electricity data, smart meter data, renewable energy information, weather data, and predictive analytics for future electricity demand forecasting.                                                                                       |